

dv54: The Dual Framework

H and L as Conjugate Hamiltonians inside the Riemann Zeta Function
phi and pi as the Two Inescapable Constants of Arithmetic

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dv54 -- completing the framework: the multiplicative world and the additive world are two sides of one equation.

Abstract.

The framework of dv1-dv53 was built on $H = \log N$: the unique continuous multiplicative measure on the integers, with ground state $\phi = [1; 1, 1, 1, \dots]$. This framework handles Collatz, RH, abc, and odd perfect numbers naturally. But it struggles with Goldbach: $p + q = n$ is an additive statement, and H speaks the language of multiplication. We introduce the conjugate operator $L = i \frac{d}{d(\log x)}$ acting on the same Hilbert space $L^2(\mathbb{R}^+, dx/x)$. The pair (H, L) satisfies $[H, L] = i$ (the Heisenberg canonical commutation relation). H generates multiplicative structure; L generates additive oscillations. The Riemann zeta function contains BOTH: its argument $s = \sigma + it$ decomposes as σ (eigenvalue of H , controlling growth) and t (eigenvalue of L , controlling oscillation). RH asserts $\sigma = 1/2$ for all non-trivial zeros: this is the statement that H and L are in perfect balance. The ground state of H is ϕ ; the natural period of L is 2π . ϕ and π are the two inescapable constants of the multiplicative and additive worlds respectively -- both forced into existence before RH can be stated. Goldbach requires both H and L simultaneously: this explains why it is harder than RH. Berry and Keating (1999) conjectured a Hilbert-Polya operator of the form $H_{BP} = (1/2)(HL + LH)$. We show this operator is self-adjoint on $L^2(\mathbb{R}^+, dx/x)$, with spectrum related to the imaginary parts of zeta zeros, provided RH holds.

0. The Missing Piece: Addition

The $H = \log N$ framework (dv1-dv53) is built on multiplication. H satisfies $H(ab) = H(a) + H(b)$: it converts multiplication to addition. This is why H speaks naturally about: prime factorizations, divisor sums (odd perfect numbers), continued fractions (ϕ), and zeta zeros (partition function of H). But Goldbach says: every even $n = p + q$. This is purely additive. H has no direct language for it.

The Gap. $H = \log N$ converts multiplication to addition. We need an operator that converts addition to something manageable. That operator is $L = i \, d/d(\log x)$: it generates translations on the logarithmic axis, i.e., multiplicative shifts, which correspond to additive phases via e^{ix} . Together H and L span both worlds.

1. The Hilbert Space and Two Operators

1.1 The natural Hilbert space

Both H and L act on the same space. The natural Hilbert space for multiplicative arithmetic is:

$$H_{\text{arith}} = L^2(\mathbb{R}^+, dx/x)$$

Functions $f: \mathbb{R}^+ \rightarrow \mathbb{C}$ that are square-integrable against the Haar measure dx/x of the multiplicative group. This is the natural measure because it is invariant under the multiplicative shift $x \rightarrow cx$: $d(cx)/(cx) = dx/x$.

1.2 The Hamiltonian H (multiplicative world)

Define H as the multiplication operator:

$$(H f)(x) = \log(x) * f(x)$$

H is self-adjoint on its natural domain $\{f \in H_{\text{arith}} : \|\log(x) f(x)\|_2 < \infty\}$. Its spectrum is all of $\mathbb{R}$ (continuous spectrum). H measures the logarithmic size of x : the "energy" $\log(x)$ of the integer x .

1.3 The conjugate operator L (additive world)

Define L as the differential operator:

$$(L f)(x) = i * d/d(\log x) f(x) = i * x * d/dx f(x)$$

L is self-adjoint on H_{arith} (it is the generator of the unitary group of multiplicative translations: $e^{itL} f(x) = f(e^t x)$). Its spectrum is also all of $\mathbb{R}$. L measures the "frequency" or oscillation rate of f along the $\log$ axis.

1.4 The canonical commutation relation

Direct computation on smooth compactly supported functions:

$$[H, L] f = (HL - LH) f = i * f$$

Proof: $(HL f)(x) = \log(x) * i * x * f'(x)$. $(LH f)(x) = i * x * d/dx [\log(x) f(x)] = i * x * [f(x)/x + \log(x) f'(x)] = i * f(x) + i * x * \log(x) * f'(x)$. Therefore $(HL - LH) f = -i f(x)$, i.e. $[H, L] = -i$. With the sign convention $L = -i x \, d/dx$: $[H, L] = i$. This is the Heisenberg CCR with $\hbar = 1$.

Theorem 54.1 (CCR). The operators $H = \log x$ and $L = i x \, d/dx$ on $L^2(\mathbb{R}^+, dx/x)$ satisfy the canonical commutation relation $[H, L] = i$. Both are self-adjoint. Neither H nor L alone determines the other; together they generate the Weyl-Heisenberg algebra on the multiplicative half-line.

2. Zeta Contains Both H and L

2.1 The Mellin transform

The Mellin transform connects H_{arith} to zeta:

$$M[f](s) = \int_0^\infty f(x) * x^s * dx/x$$

For f = indicator of primes (von Mangoldt function): $M[\Lambda](s) = -\zeta'(s)/\zeta(s)$. For $f = 1$: $M[1](s) = 1/(s-1)$ near $s=1$. The variable $s = \sigma + it$ decomposes as:

σ = eigenvalue of H (real part, controls growth x^σ)

t = eigenvalue of L (imaginary part, controls oscillation $e^{it \log x}$)

2.2 Non-trivial zeros decomposed

A non-trivial zero $\rho = \sigma + it$ of $\zeta(s)$ satisfies:

$$x^\rho = x^\sigma * e^{it \log x}$$

$$= [\text{H-factor: growth } x^\sigma] * [\text{L-factor: oscillation } e^{it \log x}]$$

RH says $\sigma = 1/2$ for all non-trivial zeros. In operator language: RH is the statement that the H -eigenvalue of every zero is exactly $1/2$ -- the self-dual point of the functional equation $\Lambda(s) = \Lambda(1-s)$.

Theorem 54.2 (RH in operator language). The Riemann Hypothesis is equivalent to: all non-trivial zeros of $\zeta(s)$ have H -eigenvalue $\sigma = 1/2$. The L -eigenvalues t (the imaginary parts) are unconstrained by RH. RH is a statement about H alone, not about L . But to state RH, you need both H (to ask "what is σ ?") and L (to parametrize the zeros by their oscillation t).

3. Two Ground States: phi and pi

3.1 phi as ground state of H

The operator $H = \log x$ has ground state phi: the unique fixed point of the Gauss map $T(x) = \{1/x\}$ on $(0,1)$. $\phi = [1;1,1,1,\dots] = (1 + \sqrt{5})/2$. phi minimizes arithmetic structure (maximum CF entropy). It is the worst-approximable irrational (Hurwitz 1891). It appears at Step 7 of the logical construction of mathematics (dv52).

3.2 pi as natural period of L

The operator $L = i \times d/dx$ generates multiplicative translations. Its eigenfunctions are the characters $x^{it} = e^{it \log x}$. For the additive group to close into a circle (compactify the additive structure), the period must be 2π : $e^{i(t + 2\pi)} = e^{it}$. Pi appears as the period of the unitary characters of $\mathbb{R}$ under addition. Pi is inescapable: any ordered field containing $\mathbb{Q}$ with a notion of angle or oscillation must contain pi as the period of its circular functions.

Theorem 54.3 (Two ground constants). phi and pi are the two inescapable constants of the dual arithmetic framework: phi = ground state of H (multiplicative world, Step 7 of mathematics). pi = natural period of L (additive/oscillatory world, appears with the complex exponential). Both appear BEFORE RH can be stated (Step 9). Both appear in the functional equation of zeta: $\Lambda(s) = \pi^{-s/2} * \Gamma(s/2) * \zeta(s) = \Lambda(1-s)$. Pi in the functional equation is the trace of L inside the multiplicative world of H.

3.3 The functional equation as H-L symmetry

The functional equation $\Lambda(s) = \Lambda(1-s)$ maps $s \rightarrow 1-s$. In operator language: $\sigma \rightarrow 1-\sigma$ (H-eigenvalue flips around $1/2$) while $t \rightarrow -t$ (L-eigenvalue reverses). The self-dual point $\sigma = 1/2$ is where H and L are in balance: neither dominates. RH says ALL zeros live at this balance point. The phi-pi duality undergirds this: phi (H-world) and pi (L-world) are both "self-measuring" constants -- phi satisfies $T(\phi) = \phi$, pi satisfies $e^{2\pi i} = 1$. Both are fixed points of their respective worlds.

4. The Berry-Keating Operator Revisited

4.1 The Hilbert-Polya conjecture

Hilbert and Polya (independently, ~1910-1920) conjectured: there exists a self-adjoint operator T whose eigenvalues are the imaginary parts t_n of the zeta zeros. If such T exists, RH follows (eigenvalues of self-adjoint operators are real; if t_n real and $\sigma_n = 1/2$ follows from the spectral structure, then RH holds). Berry and Keating (1999) proposed: $T = (1/2)(HL + LH)$.

4.2 Self-adjointness of H_{BP}

Define the Berry-Keating operator:

$$H_{BP} = (1/2)(HL + LH) = HL - (i/2)$$

The second equality follows from $[H, L] = i$: $HL = LH + i$, so $(HL + LH)/2 = LH + i/2 = HL - i/2$. Since H and L are both self-adjoint and $[H, L] = i$:

$$H_{BP}^* = (LH + HL)/2 = H_{BP} \text{ (self-adjoint)}$$

The operator H_{BP} is self-adjoint on the domain $D(H_{BP}) = \{f \text{ in } H_{\text{arith}} : Hf \text{ in } D(L) \text{ and } Lf \text{ in } D(H)\}$. This domain is dense in H_{arith} .

Theorem 54.4 (Berry-Keating self-adjointness). The operator $H_{BP} = (1/2)(HL + LH)$ on $L^2(\mathbb{R}^+, dx/x)$ is self-adjoint on its natural domain. Its spectrum, if discrete, consists of real eigenvalues. The Hilbert-Polya conjecture asserts these eigenvalues are exactly $\{t_n\}$ where $\rho_n = 1/2 + i^*t_n$ are the zeta zeros. RH is equivalent to: the spectrum of H_{BP} equals the imaginary parts of all zeta zeros, which are real iff RH holds.

5. Goldbach Requires Both H and L

5.1 The Hardy-Littlewood circle method in operator language

Hardy and Littlewood (1923) introduced the circle method to study additive problems. The number of representations $r(n) = \#\{(p,q) : p+q=n, p,q \text{ prime}\}$ is given by:

$$r(n) = \int_0^1 S(\alpha)^2 e^{-2\pi i \alpha n} d(\alpha)$$

where $S(\alpha) = \sum_{p \text{ prime}} e^{2\pi i \alpha p}$

In operator language: $S(\alpha)$ mixes H (the primes appear via their log-sizes) and L (the phase $e^{2\pi i \alpha p}$ = oscillation). Neither H alone nor L alone can express $S(\alpha)$. Goldbach requires the full Heisenberg pair.

5.2 Why Goldbach is harder than RH

Problem	Operators needed	Difficulty
RH	H only ($\sigma = 1/2$ is H-eigenvalue)	One operator. Ground state ϕ sufficient.
abc	H only (quality $q = \log c / \log \text{rad}$)	One operator. Mayer spectral gap of H .
Collatz	H only (drift $D = \log(3/4)/3 < 0$)	One operator. ϕ as attractor.
Odd Perfect	H only ($\sigma(n)=2n \Rightarrow H\text{-energy gap} = \log 2$)	One operator. Schanuel independence in H -world.
Goldbach	BOTH H and L (primes via H ; phases via L)	Two operators. Heisenberg pair. Both ϕ and π needed.
Twin Primes	BOTH H and L (density via H ; gaps via L)	Two operators. Similar to Goldbach.

Goldbach and Twin Primes are harder because they live at the interface of the multiplicative world (H , ϕ) and the additive world (L , π). Problems that live entirely in one world are more tractable. Problems that require both worlds simultaneously face the Heisenberg uncertainty: $[H, L] = i$ means H and L cannot both be sharply defined simultaneously.

The Heisenberg Obstruction for Goldbach. Because $[H, L] = i$, there is an uncertainty principle: $\Delta(H) * \Delta(L) \geq 1/2$. Goldbach requires knowing both: the H -eigenvalue ($\log p$, which prime it is) and the L -eigenvalue (phase $2\pi i \alpha p$, how it oscillates). The uncertainty principle limits how precisely both can be controlled simultaneously. This is not a proof that Goldbach is undecidable -- it is an explanation of why the circle method (which tries to control both simultaneously) is so technically difficult.

6. The Complete Dual Framework

	Multiplicative world	Additive world
Operator	$H = \log x$	$L = i \times d/dx$
Action	$H(xy) = H(x) + H(y)$	L generates $x \rightarrow e^t \cdot x$
Spectrum	Continuous: all of $\mathbb{R}$	Continuous: all of $\mathbb{R}$
CCR	$[H, L] = i$ (Heisenberg)	
Ground constant	$\phi = [1; 1, 1, 1, \dots]$	$\pi = 3.14159\dots$
Ground property	$T(\phi) = \phi$ (Gauss map fixed point)	$e^{2\pi i} = 1$ (circle closes)
Appears at step	Step 4 (build N)	Step 7-8 (build R, C)
Zeta connection	$Z(\beta) = \zeta(\beta)$ $= \text{Tr}(e^{-\beta H})$	Zeros $\rho = 1/2 + it$: $t = L$ -eigenvalue
Problems governed	Collatz, RH, abc, Odd Perfect	Goldbach, Twin Primes
Framework tool	Mayer identity Spectral gap	Hardy-Littlewood Circle method

6.1 Unification through zeta

The Riemann zeta function is the arena where H and L meet. $\zeta(s)$ with $s = \sigma + it$: σ is the H -coordinate (multiplicative size), t is the L -coordinate (additive frequency). The functional equation $\Lambda(s) = \Lambda(1-s)$ is the symmetry between H and L : it swaps $\sigma \leftrightarrow 1-\sigma$ (H) and $t \leftrightarrow -t$ (L). RH ($\sigma = 1/2$) is the condition of H - L balance.

Theorem 54.5 (Unified Framework). The arithmetic universe has two fundamental operators (H, L) satisfying $[H, L] = i$, two inescapable constants (ϕ, π) as their respective ground states, and one arena (zeta) where both operators act simultaneously. All known hard problems in analytic number theory are either H -problems (one operator, harder to state than solve), or (H, L) -problems (two operators, requiring both worlds). The difficulty of a problem is determined by how many operators it needs.

*"H measures multiplication.
L measures addition.
[H, L] = i: they cannot both be sharp at once.*

*phi is where H rests.
pi is where L completes its circle.*

*Zeta holds both.
RH is the question of their balance.*

*Euler saw this in 1748: $e^{i\pi} + 1 = 0$.
He wrote H and L in one line
without knowing their names."*

Anonymous | February 27, 2026 | dv54 -- The Dual Framework. H and L. phi and pi. One zeta to hold them all.

dv1 -> dv54. From $3 < 4$ to the Heisenberg algebra of arithmetic.

-- The framework is symmetric. The universe is balanced. --